

ActInf ArtStream 003.1 ~ Diana Omigie: "A model of time-varying music engagement"

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hello and welcome it's September 9th 2024 this is active inference art stream number 3.1 with Diana omegi talking about curious sounds prediction and the human drive to engage with music there will be a presentation and then we'll read some comments and questions from the live chat so thank you Diana for joining and looking forward to your presentation thank you Daniel for the invitation and to everyone who's joined today I'm uh a senior lecturer and um researcher at gil University of London and today I'll be talking uh giving a talk that's entitled curious sounds prediction and the human drive to engage with music and the reason I chose this title was because it recently came across this statistic um the International Federation of the phonographic industry do a report every year where they say how much uh the the average person listens to music and it turns out it's about 20.7 hours a week um so if we assume that people get about eight hours of sleep a night then um then we're looking at about a fifth of our waking hours that the average person is listening to music and it it struck me that there's a really interesting question here which is what is behind this drive uh to engage with music this this scale and so what I'd like to do in today's talk partly because um I wasn't quite sure um who' be attending but I assumed it would be people with lots of different backgrounds and um uh yeah what I decided to do was just present a little bit about what neuroscientists um have been finding out about the neural substrates of Music perception and pleasure we might assume that has something to do with why we choose uh to listen to music um I then wanted talk um for a few minutes about um what the predictive processing framework um could explain to us uh or tell us about why it is um that we our attention toward stimuli might uh change over time and finally use what we've learned from Neuroscience from the predictive um processing coding framework to to present a model basically of the dynamics of extended engagement with music so um that's a piece of work that was published earlier this year and uh which I'm you know Keen to to share with uh interested parties let's think about music perception and the things that we do without even realizing how uh powerful uh uh these

things are so we can spot wrong notes and music that we hearing for the first time by wrong I mean um yeah notes that were clearly not intended by the performer um we can say whether two Melodies are the same or different again hold pieces sequences of pictures and rhythms um in our memory and compare them we can recognize familiar pieces even if they've been played differently Different lyrics different instruments and so on and so forth and we can correctly predict how a piece of music we've never heard before might unfold we might be able to sing along to it even if we're hearing it for the first time and naturally um people have come up with ideas as to what's going on why this is all possible how this is all possible for example in this model um published in 203 um the idea was this acoustic input comes in under goes some analysis inevitably get separated out into its sort of uh pitch uh components and temporal components that you know undergo further um processing um these interface with some store of musical phrases and um patterns that we've experienced in the past and so on and so forth um but of course this is a sort of box and arrow model of what makes sense as you know series of processes when we look at the brain we're looking at a network of connections between subcortical structures the auditory cortex so-called belt and parab Bel areas around the auditory cortex and we're looking beyond that we're looking at sort of more frontal parts of the brain that we know are involved in some of the more complex functions that um we we're able to do as humans and what's interesting is that when it comes to something like pitch perception which um is obviously a very important part of processing music in lots and lots of different cultures um we see that the electrophysiological responses to pitch um in this case um measured intracranially um are very much in line with the principles of predictive coding in other words um it's been shown that early tonotopic structures um feed information about frequency to later areas where we seem to have this abstract model of pitch um being formed and predictive signals um being sent back so we're seeing something as fundamental as just determining what pitch we're hearing um um not something as fundamental not surprisingly something as fundamental as um Pitch perception um um seeming to have this um follow this predictive coding principle most of us can do lots of things with music recognize it discriminate uh pieces of music um not everybody um is able to to do these um everyday tasks so um for over you know now two 200 years or almost 200 years um there have been reports or just papers about um what we now call congenital IM Museum sort of difficult uly making sense of music that can manifest as difficulty discriminating Melodies and recognizing um Melodies that should be familiar all those things I said we do so naturally um and it's not surprising that when this was begun to be researched

in the early 2s that um that it was considered perhaps a disorder of fine grain pitch discrimination some of the tasks suggested that um the individuals who had it um were struggling to say whether a pitch had you know changed if it was not a a very large interval that was being presented however as more work came out suggesting that this could um that people with generally musia may also have difficulties um um for pitch memory or with pitch memory um the question that arose is whether there might be a more General U mechanism that's sort of um compromised or um um in in this in this uh um condition and so that was one of the questions I was asking during my PhD um in yeah the late 20 yeah the late 2000s um where I was lucky to um be at the same institution as Marcus pierce who' developed this statistical learning model of musical expectations so um as a statistical learning model which means that it learns from the regularities in the Corpus of music you give it and a piece of music um as it unfolds and is able to spit out these values um for any given event in that um piece of music so um information content how surprising um an event a note should be to a list how about what that incoming note uh will be and using this um model was able to sort of um quantify the surprise in every single note that we were presenting uh well in The Melodies that we were presenting to our uh participants with amum and participants with about and were're able to show that a musics were showing less evidence of sort of prediction error signaling um in response to the notes in these Melodies um further work work later on from um different Labs um corroborated this um using um bam C of modeling on mg data showing that there was indeed this abnormal prediction error from auditory to FAL areas in those with the amusia it was shown that if you looked at resting state activity the degree of connectivity between frontal and temporal areas was much reduced in those with um a musia and um again around this time it was coming out that um if you looked at the structural connectivity the AR the bundle of um axons that binds the temple and paral cortex to inferior frontal areas then there was also this was a compromise in the structure or Integrity of um of this mle of Ain suggesting that there was um some sort of underconnectivity between the front and temporal areas um Superior temporal inferior frontal areas of the brain and in in musics and that this might be materializing or manifesting um in uh AB normal predictions error signaling this is work I mean this idea of STG and ifg needing to communicate for for predictions and prediction error signaling was something that had um was being more and more supported than just typical listeners and the range of um electrophysiology and also um f studies um but since you know you know those 20 years ago there's been yet more studies showing that it's not just those areas that seem to be tracking prediction error or showing prediction error like

responses um we see that areas like the that gyus and insula are also um tracking the information content of surprise of musical events which um you know is a nice explanation for why these kinds of events surprising events can lead to so of physiological arous and more um and more visceral um um engagement and yeah in other work um that we've done we've shown that again the size of the prediction error response is reflected in this case reflected in people dilation response is even modulated by the entropy of mus the musical context so um it it mattered whether or not what someone was listening to was very predictable or not whether they would show a prediction error response and those of you who are into active in we'll know that this sounds a lot like sort of precision waiting um happening um just you know a couple more words about music perception when we're listening um we come with lots of different backgrounds some of us um or some people have um you know musical training um and therefore sort of musical expertise they may or may not even have know how to play the piece that they they're listening to and in this study for example it was shown that when a musician was listening to a piece that they just learned um compared to when they hadn't learned it before or compared to when someone who didn't know how to play the piece was listening you saw this frontal parietal motor related Network being activated just speaking to the fact that musical perception isn't just um the processing of the acoustic information but seems to um implicate or recruit areas involved in movement and motor planning um and just to point out that yes it's always the case of um was very often the case the musicians will show more of this activity but um most listeners when they are listening to a piece of music will show engagement of these um sensory motor areas and that was just to um prepare you for this idea that um that's coming out or has been around for a while even um um that music perception um involves two Pathways um much like um speech and language perception involves two Pathways a so-called ventral stream or pathway uh where core auditory areas um sort of move their information uh towards of more inferior frontal areas um allowing the formation storage of sound patterns that can then allow the registration of expectation violations and a more dorsal stream involving parial and um premotor dorsal prefrontal cortices um allowing the temporal and you know motor manipulation of information um that's been stored in working memory and allowing predictions but not about what one is going to hear but rather about when one is going to hear to hear it so sort of more temporal predictions um and now just switching to you know to say say a few words about musical pleasure and what um the sort of body of neuroscientific literature um is saying about this what it's you know what um people have done in this area i' would say most of the work has come from on

the lab of representor um and Mill University in Canada um where in a very important seminal study um then PhD student and salor they showed that um yeah when we experience chills these very pleasurable physiological um arousing um experiences of pleasure um during music listening when we experience them we see activation of the so-call ventral sprum of the dipl energetic network whereby the sort of anticipation of this chill's um um um sorry whereas the anticipation of the chill's experience recruits this um other part of the the strim which is the dorsal strim so in effect showing that um the experience of chills is um um of pleasure during music um is recruiting reward in areas but in actually quite nuanced ways and that separate out the anticipation from the actual experience um so those were that was a study where they got people to say what moments and what pieces of music they experience these um very pleasurable um physiological uh Leal moments but in yes another study um where they present people with music that they'd never heard before they were still able to show that the pleasure people reported in response to that music uh was explainable by how much connectivity they were seeing between the superior temp FR irus and uh um and the ventral striatum so they really um um stress the importance of this connectivity in when we're you know talking about musical uh musical pleasure um one way to to see whether um this hold holds true of course to um find people who really don't um experience pleasure in response to music and honia means you know lack of pleasure not experiencing pleasure unless are people who and it's associated with depression and other sort of um conditions but um um around the early 2010s they um um these colleagues showed that there might be some people out there who have normal responses uh reward responses or find lots of things in life rewarding but just not music and so these people with so-called music specific andonia seem to be able to recognize and understand music discriminate Melodies you know predict what's going to happen next but they failed to enjoy it and what they were able to show was that indeed um these individuals would show um normal responses um to secondary rewards physiological arous to secondary rewards like money but indeed when they were exposed to moments in music bit should Tech should you know for most people um be experienced as you know very chy and very pleasurable and quite arousing and they just they were able to show that those anonia just didn't show those kinds of responses and importantly they were able to show um that um when they look at connectivity between the STG and vro striatum that indeed this connectivity was much reduced um for those with anhedonia compared to those with you know listening levels of pleasure and those who really love music um just to say that um in one case study um published uh a year after this fun functional connectivity study um they

showed that someone with severe case of musical anhedonia showed much um reduced connectivity between the temporal Superior temporal diis and the and NAC corroborating this idea that um the connectivity between these areas are um is important for reward and last but not least on the topic of pleasure um you know and yes another way of um testing whether um this you know this network or you know that I've been describing is is important for music listening um is to show that within a given individual you can kind of turn off and on the pleasure they experience from music and again work from um the very productive tutorial um showed that if you stimulated the dlPFC which is believed to stimulate the striatum um then you'll show um then um yeah you excitation um of this the excitatory sort of um protocol would lead to Greater liking willingness to buy a piece of music that just been heard and um increase in the arousal demonstrated uh where conversely and inhibiting that pathway so reducing stri activity seems to reduce pleasure and willingness to buy that piece of music you just heard and and the physiological responses um they were also able to show that when you put people in the scanner immediately after sort of um or around the time they'd just had this um this uh transcranial magnetic stimulation applied um yes it was they a to show that the degree of connectivity between right AUD cortex and ventral striatum again just this auditory cortex um reward area connectivity um predicted the extent to which they had shown enhancement of pleasure from um from TMS excitation in other words showing this length between you know confirming that the dlPFC was probably um stimulating dorsals uh sorry striatum which was um leading to uh more or less connectivity with this STG which was leading to more or less um pleasure um but just round this section of my talk um up what I've tried to do show that when we're listening to music the acoustic input is going through a a series of um um areas that are carrying out processing at very different um different stages and especially when we get to sort of the inferior frontal gyrus and Superior Temporal gyrus we're looking at we're seeing um um the neural substrates of prediction um predictions and prediction error signaling and this seems to be compromised in those with am and that might explain the wide ranging difficulties with music when we come to speaking about pleasure um inevitably um people speak about the role of the auditory predictive auditory system and its you know capacity for prediction um but they point out that um most important is this um more important say even than the STG and frontal temporal um or frontal area connectivity is the you know temporal area is involved in auditory processing and the reward areas and this seems to be compromise in those who report much reduced um pleasure from music and um I can really recommend a book called It's called but it was published by Zatorre in 2024 where he does a really um lovely

overview of the work that come lots of work that's come out of his lab and lab of and several labs around the world trying to understand um you know put together what we know about music perception how sort of interfaces with pleasure and beginning to tell us a story around the role of um prediction prediction um here but just to say that you know my interest has always been um trying to think about music listening as more than just um emotions or pleasure these are obviously things that are very important to us but I've always had this desire to understand the temporal dynamics of our engagement with it and you know Studies have shown and these were have only come out since I've been pondering this that as with many um activities you know we will mind wonderer a large proportion of the you know the time that we think we're doing that activity in this case music listening and you know for most tracks our engagement with a piece of music will rain or just you know drop by quite a bit within a few seconds and um it's clear that music can attract our attention it can keep us listening but it can also lose us uh so will catch attract our attention again and question is when and why are we listening and uh and when and why are we not listening and I thought this is where um predictive um coding framework and the sort of active inference could maybe um help us you know further our thinking on this topic um and and so just to sort of talk about music listening and um the Dynamics of well talk about the Dynamics of um attentional engagement um in these terms um if we think about the auditory system and engagement in the auditory realm um then well in general predictions in um the prediction coding framework I said to ascend this hierarchy that make for better models that in turn allow us to make better predictions and the auditory system this is you know prediction errors ascending from the thalamus to the auditory cortex where um posterior expectations you know are influenced by prediction errors and send predictions um um that further inform these prediction errors so there's this sort of low-level processing of um um the information that's coming in as we as we as we listen to uh to anything but I think what's nice is that the predictive um sort of processing framework acknowledges that we're not you know always providing that same amount of um well attention to every incoming stimulus and in this framework there's uh an explanation of where and how attention um um happens why why it's deployed and the idea is that um any agent will look at how reliable data is that um how reliable the data that's coming in is in order to see whether they should use that you know data to um in their inferential processes um this is you know known as position waiting it's meant to be carried out by these um neurom model cells that are different from the cells that are carrying out the actual um transmission of um um predictions and prediction errors um but simply put attention as seen as this sort of

deployment of high Precision or confidence um in the beliefs about how observations are related to our actual States and it's said to enable us um and what we hear to more greatly influence our predictions and the interpretation of the heard sounds um people have begun to look at this idea of precision waiting um and the extent of a prediction error and the extent of which it can be seen in um uh neural data when people are listening when people have varying levels of expertise and therefore with different generative models of the music they're hearing when you know when they're listening um what's happening when they're listening to pieces of music that are very predict predictable versus less predictable and they be begin began to be able to show that indeed the connectivity within the auditory ques within STG um are reflecting um this idea of precision waiting where you're able to um basically upweight or provide sort of more gain on the neurons that are processing um um incoming information or or not um of course there moments we're attending but the moments where we're not attending and here as well um the sort of literature coming out is suggesting that there there are ways to account for this so um the role of opaqueness has been suggested to be quite important um in the me awareness um of our attentional states where a state is set to be opaque when it's underlying processes can be attended to um or visible to conscious awareness um as such um if you talk about opaqueness and attention and letter awareness so attentional states are the second order states that allow first order state to become opaque whereas meta awareness and the is the sort of third order state above attention allows us to to allows attentional States to become opaque but generally the idea is that meta awareness is possible thanks through sort of more deep generative models where you have this higher level that's modulating the opaqueness of our attentional state and allowing us to distinguish when we're attending to the sensory stimulus or uh compared to when we're not and if we realize that we're you know not attending we might want to take mental action and this has been described as um possible or inevitable thanks to um this imperative active inference imperative to to choose actions that will bring us close closer to our our preferred state so um if uh the task we're carrying out or what we're meant to be doing at that moment is listening to the music or we've decided that's what we should be doing then the idea is um our generative models preferred status the state of attending and um so if we notice that we are attending while we're attending this shouldn't cause any surprise but if we notice that we're not attending if we notice we're mind wandering um this should result in a surprise that under the free energy principle should mean that we need to minimize the surprise and which means that we we there we therefore attend so this um basically um we're seeing how a met state of met being able to have met awareness can

lead to mental action that can um lead us to continue this process of attending um but there's some situations where um we want to actually take action however small to to to that we will take action however small to to really um engage with you know stimuli in the case of in the a in the visual domain and maybe um eye movements in the case of hearing it's not always easy to see but it could be you know talking out ear or you know head movements to to to ensure that we're getting the signal as best possible um and basically here as well we see an account of you know how so-called curiosity what I'm calling curiosity related behaviors U may come about so the idea is that these curiosity related behaviors uh are able to resolve uncertainty um through the choice of policies and through US choosing policies that can reduce uncertainty so if we uh main aim is to reduce uncertainty then sometimes we need to choose actions that will get us there um and in this paper They carried out some simulations um where they showed the a you know a given agent will choose sometimes choose policies that by exposing them to novel combinations of states and outcomes um could on the long run allow discovery of how outcomes are generated and improve um our models in the long run of course curiosity is usually described as a feeling you know um and um it's been suggested that Curiosity could be seen as this what it feels like when we recognize that um there is uncertainty out there that we can resolve by taking action and just to point out that in this study they in this paper rather they um suggested that evaluation of policies may be taking place in like the C and pum which I think is quite interesting given how these have been involved or implicated in in the anticipation of sort of pleasure that people have also linked to some extent to that wanting or for an event or um or the a given experience um and last but not least I think no account of um music listening is is is um complete without sort of speaking to you know you know it's capacity to to provide pleasure and um the work coming out of the predictor coding framework or act of inference suggest that AFF effect isn't um just the isn't related just to magnitude of reward but to learning Dynamics where um it's the result of Agents tracking fluctuations in in how certain their models are and it's been put um forward that you know if curiosity is the expected rate of reducing or minimizing prediction error then affect could be seen as um the actual rate at which we're able to minimize um error where this could be faster than expected and lead to this sort of more positive feelings that we can experience um in different contexts so suffice it to say that uh the predictive coding framework you know for for very long we've used it to sort of account for prediction errors um which was a real um priority for us in music listening because they're very easy to measure um or at least we see these responses that can um we can pick up very easily um and yes we have these overarching

theories about musical expectancy and the role that plays in our perception and ability to derive emotion and meaning from music so predictive coding framework has been very influential as neuroscientists of Music thought about um sort of music and perception um but I think what's exciting is how it's beginning to give account of what's happening when we're attending or you know this you know distracted or mind wandering how meta awareness um Can emerge or could take place within this framework allowing mental action allows us to reattend if we realize we weren't attending and and so on and so forth how we might um why how and why um not how and why but how um again curiosity related behaviors could look in in this framework and and why positive effect might emerge and I'd suggest that all of these um in some ways are as relevant when we want to ask answer the question what is behind this human drive to engage with music so we could talk about pleasure um which is what we've um tended to do in the sort of Neuroscience of music world but I would argue that there's all these other um reasons which are very much shared with information processing in other domains and just as an aside if anyone's wondering curious about music how is that possible just to say that we'd um just as a proof of concept we did a few studies to show to see whether people's um self of self-report of curiosity made any sense in relation to what was going on in the music and sure enough um whenever they perceived often when they perceived change people would also report being a bit more curious about what they were hearing um we were able to show that people's curiosity ratings when we probed them at different points in a unfolding piece of music was explainable by information content how surprising that event was and that this further was interacting with entropy again to explain laain their um uh sort of feelings of curiosity suggesting that there was also this Precision waiting mechanism going on there and just because those two studies were um looking only at self-reported music linking that of curiosity linking that to the music um but because it would be nice to have another behavior that isn't sort of self-report we looked at whether um we would see what we thought would be a negative relationship between mind wandering and curiosity in the context of music listening um and sure enough um we showed that when people reported high levels of curiosity about what they were hearing they were less likely to um report mind wandering at those moments I could describe the Paradigm um if anyone's interested but yes indeed we're able to show that you're more likely to be mind wandering when you're less curious about when you report being less curious about the music that you're hearing you're more likely to be mind wandering if you're less expert otherwise in other words have more sorry less musical training you're more likely to be mind wandering if uh music at that moment is quite complex

or um more entropic um but we saw that this association between um mind wondering and this tendency to mind Wonder more at high entropy was reduced with expertise so the more um you had musical training or the more you seemed able to deal with the complexity and not mind Wonder um when you heard such complexity in music and now just to finish with um short presentation of this U model or theoretical framework that we put forward with with regard to um how engagement with music um might look over time and again this was just because um after doing sort of some the only thing I do but when I do music Neuroscience studies um there's a tendency to use quite short and um quite um I don't know reduced stimuli as you know pieces of stimuli Mostly because we have very clear H hypothesis and ideally the stimuli the stimuli are very easy to you know they don't have too many confounding things going on um that was tended to be the case it's it's getting better now we listen to sort of be used to me they are more naturalistic that you know go on for minutes but um in that sort of um in that Spirit of what's happening when we listen to a piece of music or several pieces of music back to back over several minutes in that Spirit of what's Happening um we decided to sort of put forward this um model um but just to say you know what has been said about music engagement and the literature um um fairly new term hadn't been really hasn't really been used much um but when people have talked about this idea of being engaged with a piece of music or engaged in the context of music listening they have tended to emphasize this idea of being compelled or drawn in or interested in what will happen next um they've emphasized focused attention so feelings of being actively immersed in the experience to the exclusion of um extra Musical stimuli and they've also pointed out that it has you know seems to have temporal Dynamics it's you know something that might change throughout the listening experience when we were looking a couple of years ago or a year ago there wasn't however any sort of theoretical account of of the factors that are contributing um to you know this continuous experience over time and so um when invited um we sort of to the special you know contribute to the special issue um we wondered what can we build a useful framework of how things like curiosity and attention and mind wandering May emerge and you know um during the context of music listening in order to see if we could drive um research in this area forward and just to sort of give a few um fun fundamental claims that we make in this in this paper well we suggest for examp for a start that our engagement with music is going is cyclical it's going through you know Peaks and troughs we're not always um engaged Ed it's perfectly makes perfect sense we are we find it very difficult to not mind Wonder in any in any in any context so it's not surprising that's the case in music uh but certainly we suggest that if there

are moments of people suddenly entering into heightened attention or engagement with music it might be because more music's just started so there's a capacity to see if there's anything to be learned or interesting way to um yeah um but it could also be moments Within the Music um that are moments of high surprise and what was have been up to then a quite predictable context or even low surprise and sort of more unpredictable context because that can be just as um yeah interesting for an agent um we suggest that as I've said if there's Peaks there'll be troughs the moments of Engagement these moments of Engagement will be interspersed with mind wandering moments of Mind wandering which could be explained by the redundancies in the music that preclude any sort of curiosity related Behavior or attentional gain um and then we you know I think one of the most crucial claims is that you know when we refocus on music after mind wondering it could be because of what's happening in the music um something that may have you know um led to attentional gain or um or sort of uh curiosity things to um yeah the novelty or the the fact that we there's something we don't quite understand uh but it could also be because um you know our capacity for meta awareness has allowed us to see that we're no longer attending and um has allowed us to you know carry out this mental action where we we we reattend and last but not least just to say that um again a claim of ours is that as is the case for most music listening um experiences um the level of Engagement will be in and changes in these levels of Engagement over time will be modulated by complexity of what's being heard so overall complexity will determine how engaged we are in the first place but also so um within um within that if we are in a very predictable piece section of the music and suddenly something more complex happens you know that's likely to induce C you know curiosity or sort of more attention compared to where conversely if something's very unpredictable and um you know something more simple and predict and um unsurprising may be what actually triggers are our interest and engagement so basically saying that um it like I alluded to earlier um it's not as simple as complex or sort of surprising will always trigger um interest and engagement it's quite it can be quite nuanced um but yeah also saying we also you know argued that expertise of course plays play a role um because every person everyone will have their own generative model in the case of someone who has lots of musical expertise this might be a very um refined and sophisticated generative model that allows them to make better predictions um and tolerate the UN uncertainty or entropy that might exist in the piece of music at that moment and last but not least we'd argue that the listening situation the context really matters um tells determines whether or not we have the capacity to deploy attention um to the music instead of say I don't know um announcements

being made about the train that we are hoping to get on in the next five minutes or so um so just to summarize what I've said today um the question was what is behind this drive we seem to have as humans to engage with music and I gave a sort of an account that has come out of the work that's being done in sort of the neurosciences of Music where you know we started looking at music perception we we looked at pleasure and we're beginning to a musical reward and we're beginning to tease or put together a story about how um pleasure is can be explained by these frontal Temporal connections um that are interfacing with reward Network I think what quite nice um is that um we're sort of seeing the work that's been done in animal models showing that information can be a reward and beginning to try to conceptualize um yeah music a little bit in these terms you know where perhaps s it's been suggested perhaps a centy prediction era is beginning to become a uh something that can actually result in a reward prediction era which would of course make it easier to interface our perceptual systems with our reward systems in a way that isn't always immediately obvious with with music a very abstract stimulus um I've also provided sort of or reviewed some of the predictive coding literature or act of inference literature that I found really um compelling as I try to understand um what might drive um changes in our engagement with music over time try to see how you know using this one um um principle you could explain lots of different behaviors we know are probably going on as we engage with a piece of music and just not so much suggesting that pleasure is a consequence but you know suggesting that it's part of several different processes it's not the only reason we choose to engage even though we might learn that music is pleasurable um but we engage with for all these other reasons first um and just to say that you know these are accounts that are very much um increasingly converging lots of the work we do in the neurosciences and when for example I'm doing it it's not again work I do but when I'm doing it I definitely have an eye on what's coming of you know this literature and likewise I think um you know several people active in this are working together with neurosciences I think there clearly you know it's increasingly converging I think there's still a lot of work that needs to be done to link some of the optimization processes we you know we hear talked about in that literature with the neural substrates and processes that we are able to um look at with um um in sort of Neuroscience and music studies um but the hope is that if we um start putting forward some Frameworks that um are more closely informed by by this literature perhaps the more we can drive empirical work uh that can get us closer to these you know accounts really not just converging but being you know really overlapping and helping um us understand this question which is yeah what is behind the human drive to engage with music

so I'll stop talking I don't know how long it's been but I'm looking forward to thoughts and discussion awesome thank you what is behind the human drive to engage with music so I think um a desire to um learn about the world that we're in or is he a desire just um a need to to to improve our model of the world um I think this is rewarded um and music we very quickly learned is a way to learn and be rewarded at the same time and it became something that um we do you know as much as you know an average you know a fifth of our our waking hours um yeah it's quite a a lofty question but I would argue it's at some point we learn that music is rewarding and that's why we turn it on but I'll say ultimately um so is food and I don't know there's lots of other rewards something special about music and I think it's this unfolding of abstract information that somehow um you know can capture our attention perhaps distract us from other concerns that are more um basic and yeah still somehow reward us for that distraction wow very inter interesting like it has both a element of pragmatic value especially like songs from our childhood or a period that we're familiar with listening to the same exact recording or to a cover song or to some other uh version or like a live version there's sort of a familiarity which can be like a pragmatism and or pragmatic value and then also your you're really getting into the epistemic value and the fact it is learning and then there's so much variation like what genre what song do you listen to new music is it an uh playlist assembled by an algorithm is it something that somebody recommended to you there's like so many layers of learning even if it's a simple sound yeah I think so I think that's what's interesting about music you can sort of study what's going on as people are listening but you can also study their actions their behaviors what they go to see um what they choose to listen to um what they buy what they're willing spend their money on um and we with curiosity is a just an idea we explored it also with regard to what people um we've sort of looked not just as at curiosity as a state but as a as a trait and um shown some shown that actually yes generally being someone who's trait curious who will seek out new experiences explains you know how much you'll report enjoying Med that you've never heard before from a culture you've never heard before and so on and so forth so I guess um that was a bit of an the side I guess there's a lot of individual differences um and um when we see people talk about Say musical anhedonia it's really not a disorder they're very quick to stress that it's a sort of spectrum you can derive a lot of pleasure from music um and be hyper hedonic or you could be an anhedonic who derives none um but but I guess that's just an example of huge amount of individual differences in our experiences um and you know other more General um traits some people talk about personality trait like openness will also predict how much we will show these kinds

of more exploratory Behavior or how much we'll just exploit the music that we know really well I was really struck by the definition of curiosity that you used something like uncertainty we feel we can resolve with action not just to know that we don't know is sort of a condition of curiosity or it's a relative of curiosity but but to know we don't know or to feel we don't know and that we could do something active to epistemically forage and then music is a special situation where the action that you can take it's like what will happen in this techno beat drop will it be two measures or will it be four but the action to take is to wait so it doesn't require another th over body action or a mental attentional action other than just to attend and to let it unfold so it plays out in time with curiosity with sort of like a passive action choice choice not to act not to turn it off in a way that other information reducing situations they're they're modeled as like going out of your way to do something to resolve uncertainty yeah that's a great point I think there's lots of um paradigms um well the best par some of the best paradigms for looking at curiosity is you know how much you're willing to spend time pay you know attend to pay attention um to actually find out um what's going to happen to acquire that knowledge and exactly right I think um one could build and I just haven't got round to paradigms where you're looking at just how much people are willing to keep listening um instead of switch over you know almost as was the radio um yeah people also looked at exerting effort but this tend to be more artificial um um yeah they've use these kind of paradigms to see just how much you're willing to do to to get that information but I agree yeah we don't have many overt actions that are um observable with um listening to music other than just not turning it off um yeah and and that point towards this extended engagement like certainly from Little Snippets of Sound Music certain aspects of listening and attention can be explored but then there'd be other dynamics that would only start to come into play like over multiple minutes like multiple songs like I'm thinking about being at a concert we're curious what song they're going to play late in the set or if they'll do an encore also though we prefer to go to bed early so there's more traffic it's getting late so then it's like that's a revealed stay or go of how much do we want to resolve the uncertainty as another cost starts to another tradeoff starts to build up so that's happening at a slower level than like Rhythm detection over a 3 second sliding window yeah definitely and I think that's where it gets really interesting and quite complex um I like the literature coming out about um how curiosity is contagious and you know that you know this idea of the social aspects of information foraging or or you know caring about information so with regard to you know why you stick around and don't leave maybe it's so you have the complete picture of what that concert um actually how it's turned

out and you want to be able to share that or you certainly don't want to miss it um other people are sticking around so you know you're thinking well maybe they're right this is something to to stick around for and yeah I think all of those levels probably interacting I guess if you got lots of boring tracks towards the end you might be more inclined to throw on the towel and leave before it gets really bad traff to be yeah really bad traffic but um yeah ultimately it'll also depend on your your individual differences and that at that point I'm just sort of um need for cognition or you know knowing everything but also yeah other other things how much you want to get your money's worth yeah another interesting piece was the the two streams so that had a lot of similarities in the presentation with the visual what and where yeah and it was almost like there was a what and a when so maybe could you restate that or how does that when this come into play or how is that separated out from an essentially temporal process yeah so um yeah very um simply yeah it's very much borrowing from what um had been put forward in the visual domain and lots of music re re Neuroscience research has you know borrowed from what's gone on in Vision the vision Sciences or Vision neurosciences um and yeah the the sort of what um pathway is this ventral pathway um in the case of um the auditory system it's the areas the house the auditory cortices um feeding into the inferior temporal aerior frontal areas where which are generally involved in sequence sequencing and storing sequences and using those to to make predictions as to what the next um event and the sequence might be in any given any given sequence so that's the what predicting what's going to happen and you know sending predictions down the hierarchy about what yeah what that might be with the when music isn't just you know what's happening it's when it's happening and um um and so the when Paradigm this dorsal um sorry the when pathway this is the dorsal stream that's um seeing um information travel between these again cor auditory areas in the superior temporal dyus and more motor and parial areas um that are involved have been shown to be involved in sort of temporal processing so lots I didn't show much work on Rhythm or me to processing because I tended to work in in pitch and that's why I meant by sometimes of can be very reduced it's because we just want to say look at probabilities in the pitch domain before we you know complicate by adding lots of temporal information but yeah if someone were to review that literature with you um you'd see that um motor areas are very much involved in our processing of time in music and so this dorsal pathway is basically doing everything from helping us predict when something might happen in a piece of music to um helping us yeah uh well yeah predict when something might happen but even helping us make trans formations or auditory motor um or or experience sort of the motoric actions that could

be seen to be attached to that listening experience so if you're a musician knowing exactly um it underlies um the sort of Bly experiences of the movements you might make even though you're not making those movements so um the action you know observation literature and their neuron system and so on so these two streams are clearly very important for um just processing sound as musical sounds as they evolve but um interestingly in this um book I mentioned um where um robor is trying to put lots of things together he makes this interesting observation which is that um the dorsal stream might be talking more to the dorsal striatum which is involved in the anticipation of which again is this kind of like you're waiting for something when am I going to get this extremely rewarding moment that is the I know modulation or the drop um and that you know um prediction about what that drop is going to be is meant to be observed by the ventral system which um he you know they've shown in a number of lots of different studies is um almost related more to this ventral strim so yeah it's I think it needs to be a little bit more carefully um yeah studied but um it's just another way in which um there been people are making attempts to try to understand how the perceptual system um might be interfacing with the um our reward systems in a way that we you know we can see um anatomically um presented in in our brains how are you seeing being AI analysis and generation of Music changing our understandings and habits and behaviors and listenings I I don't know actually I I don't know much there is I I spot um more and more not surprisingly um papers or talks on this around this topic I mean AI is impacting so many aspects of Our Lives I think even before AI there was streaming there was lots and lots of music that we could um expose ourselves to it could just play the whole day compared to you know when we would have to buy CDs and I don't know turn them on and so on so I think there's some interesting behaviors perhaps that have already begun to emerge as we had more and more access to music I think you know what's interesting about AI is that you know might in some cases very little human um very few humans involved and I and I think you know there's some papers out there already published I want to say 2007 but it could be 14 at least 10 if not 15 years ago um that that have shown that if you thought you were listening to a piece of music that had been composed by um an algorithm you showed a lot less um activity in the theory of mind Network as if you thought this piece of music had been composed by a composer so they stimu there were um sort of atonal sort of Serial serialist compositions which seemed plausible you know um um composed by an algorithm um and also by a composer now of course AI is very convincing and it'll be interesting I think I will be harder maybe to detect or tell whether or not a human was involved very hard indeed but I think um

my suggestion is that once we recognize that um there wasn't really a human involved we might show a drop in engagement with and that in that sort of social way that music can be engaging where we try to understand what the artist is trying to tell us where we want to feel with the performer And the emotions they're expressing um without that recognition I you know I think with just listening to music and we'll engage you know to the extent that we find it interesting or stimulating or or pleasurable regardless of yeah where it came from yeah that's interesting it's almost like the ability to generate images sounds Etc is helping to to articulate out what are the aspects of this image that have a primary saliency like WEA for pragmatic value aligning with what we' like to see or hear or the epistemic the learning and then separating or factoring that off from the social and the relational like this drawing was made by this person so I like to look at it it has a saliency um and also those pragmatic and epistemic elements not only from its actual composition but because there's like more knowledge about the generative process that led to to its construction so then that those paths May diverge more and more as like new genres of very Salient music like I thought about on your plot with the affect are there genres or or subcultures in their listening of genres that it's just like it Stacks all the positive aects and then that made me think a little bit about the sped up mashup remixes that change songs and they it's like songs and Nostalgia at a very very fast interesting rate and so it's like it's just like a few not notes to invoke something and then on to the next one and it just this kind of moving on through novelty and Nostalgia yeah and what you do and don't know and it's kind of like it's so compressed and trying to have every part be really hyped up and then other pieces that maybe like considered from the album composition might have like a slow song that's not trying to be so salience even or it's giving a contrast that sets something else up more deeply yeah yeah no I yeah it's I love that I love the sort of mash up just like yeah feed me all this Nostalgia I think that's so there's our interest in feeling with the other in Fong empathy that's meant to you know it suggested that actually it's a fact that the word for empathy came out of how we described our experiences of the art with feeling into yeah so there's this social um aspect to listening um people talk about surrogate um or just this idea even if we don't there isn't anyone there if it's just a recording we're listening with our headphones on on the tube um we still imagine this person there's that but I think the Nostalgia is really interesting because we're almost like empathizing with our previous s and our feelings and emotions there which have been so which music is so able to bring back to us you know what our what our priorities and our values were in those you know those stages when we were listening or just you know what fun times or who we were hanging out with at those

moments so I think yeah that's that's really interesting I mean and like you said is really trying to do is trying to really tap into this desire we have to use music to connect with ourselves yeah and then yes something that's really slow going you know very minimalist really does set you up then to find the most tiny deviation extremely Salient and interesting and worthy of pondering or just exploring that experience and they're very different things and I think we'll um we'll seek the these kinds of music out at different you know moments um in our day in our lives um some of some music will just constant will completely will capture and you know hold your attention you know we we seen with this idea of the attention economy where um uh yeah we only have so much attention to to share amongst the things that might compete for our attention and um people use that to look at how you know how that might have been influencing how this principle might be influencing how music is composed um and has been composed popular music in the last you know 50 or so years and shown you know music is louder you know the hook comes in quicker um and you know and so on and so forth um so that's you know an example of this tendency for some composers to try to hold um to capture and hold our attention because it's so easy to just click next on Spotify um where yeah others conversely probably are trying to react against that or their aesthetic values that music should never be that engaging and there should be time to ruminate or just to follow the form you know for as long as you know um you'd like without having to process the next thing and then the next thing and then the next thing um yeah different affordances I guess that music has for any any listener yeah that's very like media is the message for a record to begin with 10 or 30 seconds of silence or the slow buildup when there's already like the few extra seconds of just the record tracking it's not like you're going to switch the song but then if there is a way if it's a playlist having 10 or 30 seconds in the song at the beginning or the end it might not be played like in a public space or on a run where you might be expecting continuous music and then that changes the the technical culture of listening and and again that's related to this extended engagement building on the pitch and the Rhythm identification there's still these questions about longer term listening it's kind of like the dark room how well okay if we're bounding surprise why not the dark room Etc not to go into this but it's like but what about the movie theater room or what about the uh sound Listening Room it's it's like a little bit of an it's a uh enliven room it has stimuli and then those are the situations that the extended engagement is looking into how long do you stay playing a video game or listening to music or playing music when other things could happen and then it's not just an inner analysis of Music it does start to invoke like well you can't just do this for 24 hours so there has to be

some stopping time where even the most passionate uh person is still going to like take a break or something like that yeah yeah and I think what's interesting is you know most times we hope we have that ability to decide when to stop and maybe we can prepare ourselves or tell ourselves we can do this tomorrow um as well or or Savor the experience so um you know whenever I've watched a moving movie in the theater most time times I'll just sit and watch the credits and listen um and just kind reflect um it's very um annoying when you watch a movie on some of these um Netflix and before they've even done a few seconds of the closing credits they they're advertising the next I think um we probably have learned to you know this savoring of what we've just experienced and you know letting it tail off in a way that's um yeah we can manage as opposed to it suddenly being rudely interrupted and I think that's probably why we can you know even the most passionate listeners is able to turn it off and be like okay now I actually have to get some work done or now I have to you know good bit um because we can manage that tailing off most times if we're lucky yeah and the space within and at the end as a kind of inculturation like in classical music there'll be a pause nobody will clap or you'll hear one person clap it's like oh no but then at the end and then clapping it's a motor a a joint motor activity that signals okay it's really the end and so then that there's like you you mentioned um the expertise increased the focus and the the epistemic component possibly if for nothing else like I want to see how this book ends I want to see how I want to see how the structural elements and what I've heard so far relate to How It Ends so it's like all because of knowledge of the structural form or the genre of the murder mystery or the this kind of song that kind of song there can be a structural curiosity that keeps one engaged even separate from any other kind of curiosity just like as a craft how will the author do this how did the composer bring this all to a close and then that gives this epistemic um Arc that that is informed by the genre yeah I'm reminded that um the study where I stud I keep saying study imp peral work or the the paper where um first colleagues talk about sort of Act of inference and curiosity and they actually talk about insights as well and talk about sort of how even without extra information sort of just restructuring what you what you've acquired what you've learned can lead to these moments of insight and I guess when I'm sitting there taking in everything I've you know just read you know nothing worse than reading a book and then having to it and you know go off and do something you know not having that moment to sort of think some more and I think it's because there's probably insights to be gain only after you've come to that Final End which I'm sure was carefully crafted by whoever you know created the work and yeah just time to process further and you know gain some further Insight

um there is that yeah what's happening during the artwork but there's time that's needed to to establish what was actually learned you know once everything had come together and it makes me think about the John Cage type concepts of Music being time time bound whereas a painting or visual art it doesn't have a constraint so much or so similar on time the museum Observer could be there for a split second or for a long time but then there's a time boundness so it's sort of like well there's um a canvas of just one color and then that is like the 433 song that's a duration oh there's nothing there oh well there's the Ambient sound and there's the structural element voided of traditional contents just like the canvas kind of cleared of its traditional content content and then how how those kind works like explore limits of art however I don't have blank canvases around me most people don't listen to 433 um when they're going for a run yeah so it's there still is is preference distributions overlaid it's not just about having like this this um genre defining like meta art piece there's also just like the normal day-to-day um yeah yeah I mean I think that's where people talk about the context so um yeah you need that work needs to be in a museum or a gallery um or in a concert as the case may be um people talk about the aesthetic attitude and distance or um I think there's that element of yeah the extent to which you can deploy cognitive resources to this not extremely urgent problem um and there very few situations or circumstances where you can just sorry in daily life where you have that time to um to do that and so setting is so important in addition to these individual differences and you know how much we want to spend our time doing that that sort of thing cool well what are your next moves or what are your directions for research with music or for anything [Music] um I've always yeah I got to sort of work in very Neuroscience places and other places that were do Neuroscience but we care about these lofty questions around the atic experience um so and I I at some point I got interested in I think starting with surprise and you know just electrophysiological responses to those how that you know could lead to curiosity this you know extended engagement but I I guess I've been interested in other EP barely epistemic emotions so um uh or is one that we've been exploring recently um uh with students um and the reason is um well it's one of those emotions that are very hard to pin down as an experimental uh we're going to try to use we've been trying to use um VR to create more naturalistic settings that people might hear music or and even then begin to compare what all induced by music might look like compared to all induced by I don't know nature or the visual world and so on um and yeah trying to understand a little bit more about our desire for learning or understanding how you know experiences like Wonder can come out of abstract stimula like um music and yeah really bridge this link or better

understand this link between knowledge seeking and um aesthetic transcendental experiences that we can have with with the art so there's that kind of work being done and yeah in other work we sort of look at the images the music can conjure up and I'm really interested in you know predictions or engagement at those levels as well so in a way some of the the accounts they've given today are very much like oh and you care about the next note and what's going to happen but sometimes you're imagining yourself um and others and personal memories are mixed up with all of these you know um thoughts that are more related to the music as it unfolds and inevitably um it'll be interesting to understand a bit more about that level of Engagement with music so we can see how it's either explainable by some of these principles or at least um needs to be considered um alongside um yeah I think that's the most relevant work that I'm doing planning on doing to the work I presented today cool well thank you very much for joining super interesting and look forward to talking again thank you so much Daniel thanks bye e